Sayuja Kute

Boston, MA 02128 | (857) 391-8142 | kute.sa@northeastern.edu

www.linkedin.com/in/SayujaKute | https://sayujakute.vercel.app

EDUCATION

Northeastern University, Boston, MA 09/2024 - Present

Master's in Analytics (Major: Evidence Based Management)

Coursework: Data Analytics, Statistics, Business Intelligence, Leadership in Analytics, Risk Management

RMD Sinhgad School of Engineering, Pune, India 06/2024

Bachelor of Engineering in Information Technology, Honors in Data Science

Relevant Coursework: Machine Learning, Data Science & Big Data Analytics, Business Analytics & Intelligence, Human Computer Interaction

TECHNICAL SKILLS

Programming & Data Analysis: Python, R, SQL, Feature Engineering, Data Manipulation & Cleaning, VBA

Business Intelligence: Power BI, Tableau, DAX, KPIs/Dashboard Development, Data Storytelling, Google Analytics

Cloud & Database Management: Snowflake, Amazon Redshift, AWS, Azure.

Tools & Processes: MS Office 365,Visio, Jira, Scrum, SDLC, Pivot Tables, Informatica IICS, Alteryx

EXPERIENCE

DeepThink Health, USA | Data Analyst 01/2026 – 06/2026

- Analyzed healthcare workflow data across cases, billing, and orders using SQL, surfacing recurring data-quality trends that were skewing operational reporting.
- Investigated discrepancies between API responses and backend records, tracing root-cause patterns across workflow stages and quantifying their impact on data reliability.
- Partnered with engineering to prioritize fixes based on data-driven findings, translating technical inconsistencies into clear recommendations for stakeholders.
- Designed reusable data-validation frameworks that scaled analysis across workflow scenarios, cutting manual review time and speeding up reporting turnaround.

Sigma Automation Controls, Pune, India | Technical Intern 04/2024 - 08/2024

- Analyzed real-time PLC and sensor data using Python (**pandas, matplotlib**) to identify irregular machine cycle durations, leading to the discovery of calibration issues that improved process **efficiency by 12%.**
- Built interactive **Power BI** dashboards using production logs and SCADA exports to monitor system load and alert thresholds, helping engineers reduce unplanned downtime across two major client installations.
- **Documented** key data processes and dashboard metrics to help the engineering team understand, maintain, and reuse the tools more effectively.

PROJECTS

Flu Season Staffing & Early-Warning System | Informatica IICS Data Pipeline 07/2026

- Built an **Informatica IICS ETL pipeline** to unify 5 fragmented CDC surveillance and mortality data sources, resolving inconsistent join keys, placeholder values, and mismatched time and geographic grain.
- Found regional flu **severity and raw visit volume told opposite stories**, revealing under-resourced states carrying 7-9x higher severity despite lower visit counts.
- Identified a **1-7 week early-warning signal** between outpatient visit peaks and mortality peaks, translating the finding into staffing recommendations for regional health leadership.

Early-Season Influenza Surveillance & Mortality Prediction 12/2025

- Analyzed **CDC influenza surveillance and mortality data (2020–2025)** to evaluate early-season predictors of end-of-season influenza deaths across New England.
- Built **regression and machine learning models** (Poisson GLM, Linear Regression, LASSO, Random Forest) using lag features and holdout validation to assess early-warning performance.
- Developed a **weekly logistic classification model** with lagged ILI and mortality signals, achieving **~85% accuracy** and supporting short-term public health risk detection

Power BI Analyst Job Simulation on Forage, PwC (Remote) 01/2025

- Built KPI-driven dashboards in **Power BI** to identify underperforming departments and visualize workforce engagement using **Snowflake**-based datasets.
- Analyzed retention and productivity metrics to surface at-risk employee segments, proposing actionable **workforce planning** improvements.
- Translated workforce trends into executive-level DEI recommendations to guide data-driven policy and hiring strategy.

ACHIEVEMENTS

Leadership & Social Impact: Certified Yoga Instructor at University Recreational Center.